

Shan-Chi Sandy Hsu

Taipei, Taiwan | +886 975378161 | milkywaysandy@gmail.com | www.linkedin.com/in/sandy-hsu-taipei |

SUMMARY

Results-driven AI Engineer and Data Scientist with over 3 years of experience developing, benchmarking, and deploying production-grade Generative AI and Machine Learning solutions in high-compliance clinical environments. Specialized in Retrieval-Augmented Generation (RAG), local LLM infrastructure deployment (Ollama/air-gapped systems), and end-to-end ML pipelines. Proven track record of architecting custom Python NLP engines, optimizing retrieval accuracy, and delivering scalable AI applications for hospital integration.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, SQL, SAS
- **AI / ML / LLMs:** Retrieval-Augmented Generation (RAG), Prompt Engineering (CoT, Few-Shot), Ollama, Gemini API, LangChain, FAISS, PyTorch, scikit-learn, NLTK
- **Data Engineering & Infrastructure:** SQLite, Air-Gapped/HPC Deployment, Windows Server, Streamlit, Vector Databases, Git

PROFESSIONAL EXPERIENCE

National Taiwan University | Taipei, Taiwan

Generative AI Engineer / Research Associate | Hybrid | Dec 2024 – Present

- Architected and deployed a clinical Retrieval-Augmented Generation (RAG) prototype using Gemini embeddings, FAISS vector index, and Streamlit to query Taiwan FDA drug inserts; currently undergoing NTUH system integration.
- Engineered a high-performance custom Python indexing and retrieval engine (bypassing LangChain) to benchmark token-chunking strategies across 1,800+ complex medical documents, optimizing search precision.
- Designed top-k retrieval (top 6 chunks) with side-by-side reference displays, adhering strictly to clinical safety and explainability guardrails requested by medical staff.
- Built a structured RAG pipeline integrating SQLite for localized QA datasets, enabling offline LLMs to synthesize relational database lookups into accurate medical answers.

Healthcare Data Scientist / Research Assistant | Remote | Apr 2023 – Dec 2024

- Collaborated asynchronously across a distributed team of 8 using Line and Zoom to self-manage and deliver ML pipelines end-to-end with weekly async updates.
- Configured secure, air-gapped local LLM infrastructure on Windows/HPC using Ollama and Llama 3 models, eliminating data exfiltration risks for sensitive Electronic Health Records (EHR).
- Implemented Llama 3 (8B) models to extract insights from unstructured ECHO reports, using strict JSON schema prompt engineering to feed downstream analytics pipelines.
- Parsed and standardized 170,000+ semi-structured hospital ECG records using complex regular expressions and NLTK NLP preprocessing pipelines into clean relational structures.
- Developed and evaluated 6 machine learning models in scikit-learn, deploying a Random Forest classifier to predict treatment-induced heart failure in lung cancer patients.

EDUCATION

University of Colorado Boulder | Master of Science in Data Science | GPA: 3.82/4.0

Key Focus: Deep Learning, Natural Language Processing, Machine Learning, System Architecture

McMaster University | Master of Health Research Methodology (Completed Coursework)

Key Focus: Clinical Trials, Health Data Analysis, Systematic Review Methods

Publications: Co-authored 4 peer-reviewed medical journal papers resulting from data analysis contributions.